Note Title 9/10/2025

COL774 | COL7341
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MAP estimate | ML estimate

Data $D = \{x^{(i)}, y^{(i)}\}_{i=1}^{m}$

$\theta_{ML} = \underset{\theta}{\arg\max}\ P(D;\theta) \equiv \underset{\theta}{\arg\max}\ P(D|\theta)$ eq

under some modeling assumptions

↳ MLE

Statistician View | Bayesian View

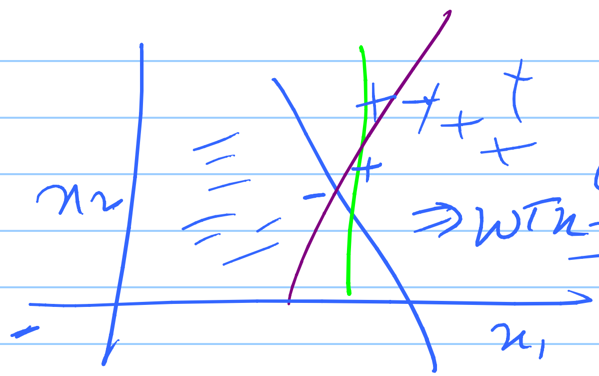
Instead of obtaining $\underset{\theta}{\arg\max}\ P(D|\theta)$ — ①

$\underline{\theta_{MAP}} \rightarrow \underset{\theta}{\arg\max}\ P(\theta|D) = \underset{\theta}{\arg\max}\ \frac{P(D|\theta)P(\theta)}{\cancel{P(D)}}$ — ②

SVM:- Support Vector Machines

(Classification)

$y \in \{-1, 1\}$



$\theta^T x = 0$

$\Rightarrow w^T x + b = 0$

$h_\theta \in$ Linear classifier (∵)

$\theta^T x = 0 \quad | \quad w^T x + b = 0$

$w \equiv (\theta_2, \theta_1) \quad b = \theta_0$

$\mathcal{J}^{h}_{w,b}$ SVM Objective:- (maximize $\underbrace{\text{minimum distance}}_{\text{margin}}$)

$w^T x + b = 0$

$\Rightarrow$ Functional Margin:- (of $x^{(i)}$)

$\left.(w^T x^{(i)} + b)\, y^{(i)}\right\} \hat{\gamma}^{(i)}$ unnormalized distance

Geometric Margin:-

$\left.\dfrac{(w^T x^{(i)} + b)\, y^{(i)}}{\|w\|}\right\} \gamma^{(i)}$ normalized distance

SVM objective:-
$$\max_{w,b} \min_{i} [\gamma^{(i)}]$$
$$\hookrightarrow \frac{(w^T x^{(i)} + b) y^{(i)}}{||w||}$$

$$\Rightarrow \max_{w,b} \gamma$$
$\forall i$:- $\gamma^{(i)} \geq \gamma$

$$\max_{w,b} \gamma$$
$$\frac{(w^T x^{(i)} + b) y^{(i)}}{||w||} \geq \gamma \Rightarrow$$

$$\left.\begin{array}{l} \max_{w,b} \gamma \\ (w^T x^{(i)} + b) y^{(i)} \geq \gamma / ||w|| \end{array}\right\} (1)$$

Let $\hat{\gamma} = \gamma ||w||$ Replace γ by $\frac{\hat{\gamma}}{||w||}$ in (1)

Then:- $\max_{w,b} \frac{\hat{\gamma}}{||w||}$

subject to $\forall i$:- $y^{(i)} (w^T x^{(i)} + b) \geq \hat{\gamma}$

Constrained optimization ~~pb~~ problem

Let $\hat{\gamma}^*, w^*, b^*$ is a solution to above problem

Then $k>0$ $k\hat{\gamma}^*, kw^*, kb^*$ is also solution

$\Rightarrow$ Assume:- $\gamma = 1$

$$\Rightarrow \max_{w,b} \frac{1}{\|w\|} = \min_{w,b} \|w\| \equiv \min_{w,b} \frac{1}{2} w^T w$$

$$y^{(i)} (w^T x^{(i)} + b) \geq 1$$

$$\boxed{\begin{array}{l} \min_{w,b} \frac{1}{2} w^T w \\ \text{st:- } y^{(i)} (w^T x^{(i)} + b) \geq 1 \end{array}}$$

SVM objective